

A Model That Always Fits

The history of adjustments to dark matter and Λ CDM, from Zwicky to the cosmic dipole — and the adjustment to come

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1. The question

There are two ways a scientific model can survive contact with new data. It can *predict* the data in advance — stick its neck out, name a number, and be proven right. Or it can *accommodate* the data after the fact — add a piece, tune a parameter, and absorb the surprise. Both happen in healthy science. The diagnostic question, the one that separates a living theory from a dying one, is which mode a model spends most of its time in.

The standard cosmological model — cold dark matter plus a cosmological constant, Λ CDM — is the most quantitatively successful description of the universe ever written down. It is also a model whose ninety-year history reads, in long stretches, as a sequence of additions and revisions made to keep pace with observations it did not foresee. This paper traces that history honestly: the genuine predictive triumphs alongside the genuine pattern of accommodation, ending with the cosmic dipole anomaly of Secrest and collaborators — a challenge that strikes not at a parameter but at a foundation — and a forecast, grounded in the track record, of how the model will most likely be adjusted again.

The point is not that Λ CDM is foolish. It is not. The point is that a model which has *never yet* met an anomaly it could not absorb is a model whose flexibility deserves examination, because a theory that can fit anything predicts nothing.

2. The birth of missing mass (1933–1980)

In 1933 Fritz Zwicky measured the velocities of galaxies in the Coma cluster and found them moving far too fast to be held together by the gravity of their visible stars. He inferred *dunkle Materie* — dark matter — and was, for forty years, largely ignored.

The idea returned in force in the 1970s, when Vera Rubin and Kent Ford measured the rotation of spiral galaxies and found the outer stars circling just as fast as the inner ones. By Newtonian gravity the speeds should fall off with distance from the bright center; instead the rotation curves were flat, far out into the dark. Independently, Ostriker, Peebles, and Yahil argued that galactic

disks would be unstable without a massive surrounding halo. By 1980 the consensus had formed: galaxies and clusters are embedded in vast halos of invisible matter, several times the mass of everything we can see.

This first step was reasonable inference, not patchwork. But notice that it was already a choice. Faced with “objects move faster than visible gravity allows,” the field chose “there is unseen mass” over “the law of gravity is incomplete at low accelerations.” Both were on the table in 1980. Only one was pursued at scale.

3. Hot becomes cold: the first major revision (early 1980s)

What is the dark matter? The first serious candidate was the neutrino — a known particle, pleasingly economical. But neutrinos are light and fast, “hot” dark matter, and fast particles stream out of small clumps before they can collapse. Hot dark matter therefore predicts that the largest structures — superclusters — form first and fragment into galaxies later, a top-down universe.

The galaxy surveys of the early 1980s showed the opposite: small things formed first and assembled into big ones, bottom-up. Hot dark matter was incompatible with the data. The response was not to question dark matter but to change its temperature: out went the hot neutrino, in came **cold** dark matter (CDM) — heavy, sluggish, hypothetical particles that clump on all scales. Blumenthal, Faber, Primack, and Rees laid out the cold picture in 1984, and it worked far better.

This was the first major revision of the dark sector driven by data rather than prediction. It was good science — the model was falsified on the hot/cold axis and corrected. But it set a template: when the dark matter conflicts with observation, adjust the dark matter.

4. Standard CDM and its crises (1980s–1990s)

Through the 1980s the default was **Standard CDM**: cold dark matter at exactly the critical density, the total matter content equal to one in the cosmologist’s units ($\Omega_m = 1$), favored because inflation predicts a spatially flat universe and matter was assumed to supply the flatness.

Standard CDM accumulated problems through the 1990s, each a quiet contradiction with data:

- **The age crisis.** With all the density in matter and a Hubble rate near observed values, the model universe came out *younger* than the oldest stars inside it — a logical impossibility.
- **The baryon catastrophe.** White, Navarro, Evrard, and Frenk showed in 1993 that galaxy clusters contain too high a fraction of ordinary (baryonic) matter to be consistent with $\Omega_m = 1$.
- **The shape problem.** The pattern of galaxy clustering implied less total matter, and a different power spectrum, than Standard CDM predicted.

The field responded not with one correction but with a *bestiary* of them: tilted CDM, mixed hot-plus-cold CDM, open low-density CDM, decaying-particle τ CDM. For most of the 1990s there was no single standard model — there was a market of adjustable variants, each tuned to relieve a different tension. This is the signature of a paradigm under strain: not collapse, but proliferation.

5. 1998: adding dark energy (SCDM $\rightarrow$ Λ CDM)

The resolution came from outside the dark-matter debate entirely. In 1998 and 1999 two teams measuring distant supernovae found that the expansion of the universe is *accelerating*. To accelerate cosmic expansion you need a component with negative pressure filling empty space — a cosmological constant, Λ , or “dark energy.”

Adding Λ was a watershed. It resolved several Standard CDM problems at a stroke: lower the matter density to $\Omega_m \approx 0.3$, let dark energy supply the remaining ≈ 0.7 and the flatness, and the age crisis, the baryon fraction, and the clustering shape all fall into line simultaneously. The new model — $\Omega_m \approx 0.3$ in matter (mostly cold dark matter), $\Omega_\Lambda \approx 0.7$ in dark energy — is Λ CDM, and the convergence of independent measurements onto it was genuinely impressive. It earned the name “concordance model.”

But step back and count. The universe now required *two* dark components, dark matter and dark energy, together about 95% of everything, neither of them ever detected in a laboratory. And the value of Λ is the worst quantitative prediction in the history of physics: the natural expectation from quantum field theory overshoots the measured value by some 120 orders of magnitude. Λ CDM did not explain dark energy. It named it, fit it, and moved on.

6. The concordance triumph — credited in full (2000s)

Honesty requires a full stop here to credit what came next, because it is the strongest part of the story and any account that skips it is propaganda.

Through the 2000s and 2010s, Λ CDM made genuine, risky, confirmed predictions. The cosmic microwave background — the relic light of the infant universe — carries a series of acoustic peaks whose heights and spacings Λ CDM predicts from six numbers. The WMAP and Planck satellites measured those peaks with extraordinary precision, and they matched. The same six-parameter model predicted the scale of baryon acoustic oscillations imprinted in the distribution of galaxies, and surveys found that ruler exactly where it should be. Big Bang nucleosynthesis, gravitational lensing, the broad growth of cosmic structure — all consistent.

This is real predictive success, not accommodation, and it is why Λ CDM is the standard model and not a curiosity. A fair critic concedes it without flinching. The interesting question is what the *same* model did when it left the clean, linear, early-universe regime where it shines and entered the messy, nonlinear regime of galaxies — and what it is doing now, as a new generation of data arrives.

7. The small-scale crisis and the feedback patches (2000s–2010s)

Inside galaxies, cold dark matter’s clean predictions began to fail:

- **Missing satellites.** CDM simulations predict thousands of small dark subhalos around a galaxy like the Milky Way; only a few dozen luminous satellites are seen.
- **Cusp versus core.** CDM halos should rise to a dense “cusp” at the center; many real dwarf galaxies have flat “cores.”
- **Too big to fail.** The biggest predicted subhalos are too dense to match the observed satellites — they should have formed visible galaxies, and didn’t (Boylan-Kolchin and colleagues, 2011).

- **Planes of satellites.** The satellites of the Milky Way and Andromeda lie in thin, rotating planes, which randomly accreted dark subhalos should not produce.
- **The radial acceleration relation.** Galaxies obey a remarkably tight law (McGaugh, Lelli, and Schombert, 2016): the gravity you observe is fixed entirely by the visible matter, with a single transition acceleration appearing in every galaxy. Dark matter, which need not track visible matter, must arrange itself to reproduce this relation galaxy by galaxy.

The responses were, once again, additions rather than predictions. Where the dark matter failed, astrophysical *feedback* was invoked — supernova winds, black-hole outflows — and tuned inside ever more elaborate simulations until the small-scale data were reproduced. Where feedback strained, the dark particle itself was modified: warm dark matter, self-interacting dark matter, fuzzy ultralight dark matter. Each anomaly admitted a fix; each fix added a parameter or a process. None was forced in advance by the theory; all were calibrated after the fact to the thing they were meant to explain.

8. The tensions era (2010s–2020s)

As precision improved, the accommodations stopped being quiet. Several now rise to the level of formal *tensions* — disagreements between independent measurements that the single model cannot reconcile without new ingredients:

- **The Hubble tension.** The present expansion rate inferred from the CMB (Planck, $H_0 \approx 67.4$) disagrees with the rate measured directly from the local distance ladder (SH0ES, $H_0 \approx 73$) at more than five standard deviations. The proposed cures are, predictably, additions: *early dark energy* switched on before recombination, extra relativistic species, modified recombination physics.
- **The S_8 tension.** Surveys of gravitational lensing and clusters find the universe slightly *less* clumpy than the CMB-calibrated Λ CDM predicts, a two-to-three-sigma mismatch in the growth of structure.
- **JWST’s early galaxies.** Beginning in 2022, the James Webb Space Telescope found galaxies at very high redshift that look more massive and more mature than Λ CDM’s timetable for structure formation comfortably allows — a tension still being digested, with the usual menu of fixes (revised star-formation efficiency, new physics) on offer.
- **DESI and evolving dark energy.** Most strikingly, the Dark Energy Spectroscopic Instrument’s measurements of baryon acoustic oscillations, released in 2024 and 2025, prefer a dark energy whose strength *changes with time* ($w \neq -1$) over a constant Λ , at a significance that climbs toward four sigma in some dataset combinations. If this holds, the “ Λ ” in Λ CDM — the one component the model treats as a fixed constant — is not constant at all, and the model must absorb a *dynamical* dark sector.

The shape of the era is clear. The model is not being abandoned; it is being extended. Each tension is met by promoting a constant to a function, or adding a field that switches on at the right moment, or invoking a process tuned to the gap. This is exactly the behavior of section 4’s Standard CDM crisis, replayed at higher precision.

9. The cosmic dipole: a challenge to a foundation (2021–present)

The deepest of the recent challenges does not touch a parameter at all. It touches an *assumption*.

Λ CDM is built on the **cosmological principle**: that the universe, viewed on large enough scales, is the same in every direction (isotropic) and at every place (homogeneous). This is not a measured fact so much as a foundational simplification — the assumption that makes the whole mathematical edifice (the Friedmann–Lemaître–Robertson–Walker framework) tractable. Nearly everything in cosmology rests on it.

There is one clean way to test it. As the Solar System moves through the universe, our motion should make the distant sky look slightly brighter and more crowded in the direction we are heading and dimmer behind us — a “dipole.” The cosmic microwave background shows exactly such a dipole, and from it we infer our velocity, about 370 kilometers per second. The cosmological principle then makes a sharp prediction: the dipole measured in *any* sample of distant sources, from our motion alone, must match the one the CMB gives. Same direction, same amplitude. No freedom.

Nathan Secrest and collaborators tested this. In 2021, using 1.36 million quasars from the space-based WISE survey — chosen precisely because, being in space, it avoids the ground-based systematics that plague radio catalogs — they measured the quasar dipole. It pointed in roughly the expected direction, but its amplitude was **more than twice** what our CMB-inferred motion predicts, rejecting the purely-kinematic interpretation with a p-value of about $5 \times 10^{-7} - 4.9\sigma$. In 2022, combining the WISE quasars with NVSS radio galaxies — an independent dataset — they tightened the result to 5.1σ , and titled the paper, without hedging, *A Challenge to the Standard Cosmological Model*.

Read plainly: either we are moving about twice as fast as the CMB says (which conflicts with the CMB itself), or the distant universe is genuinely *not* isotropic — and the cosmological principle, the assumption underneath Λ CDM, is wrong at a level that matters. Unlike the Hubble or S_8 tensions, this cannot be patched by adjusting how much dark energy or dark matter there is. It questions whether the symmetry the whole model is built on actually holds.

Honesty requires the other half. The anomaly is real and persistent across independent catalogs, but it is also under vigorous reassessment. As of late 2025, new analyses — Bayesian re-estimators, reworked CatWISE samples, careful accounting of the kinematic contribution to source counts and of flux and redshift-evolution effects — are actively probing whether part or all of the excess is an unmodeled systematic or selection effect rather than a true cosmic anisotropy. The dipole is not a settled overthrow of Λ CDM. It is a sharp, foundational anomaly whose verdict is still being argued. That is exactly the moment worth watching.

10. The adjustment to come

What happens next is, on the historical evidence, easy to forecast in *shape* if not in detail. Λ CDM will most likely be adjusted again, and the adjustment will most likely take one of the field’s three customary forms:

1. **Promote a constant to a function.** This is already underway with dark energy. If DESI’s preference for evolving dark energy survives — and the larger DR3 and successor datasets will decide within a few years — the constant Λ becomes a time-dependent field, $w(z)$, and the

model becomes $w_0 w_a$ CDM, with two new parameters where there was one fixed number. The cosmological-constant problem is not solved by this; it is relocated.

2. **Add a component that switches on at the right time.** This is the template for the Hubble tension: an *early dark energy* that appears briefly before recombination, raises the inferred expansion rate, and then disappears — engineered to close a specific gap and otherwise leave the successful predictions intact.
3. **Weaken a foundational assumption locally.** This is what the dipole will demand if it survives scrutiny: some controlled departure from exact isotropy — a “tilted” universe, a very-large-scale structure, a superhorizon mode, an anisotropic expansion — added with enough freedom to fit the observed dipole while preserving the CMB’s near-perfect isotropy. The harder the dipole is to dismiss as systematics, the more such machinery will be required.

In every case the model survives by gaining a part. This is not a prediction that Λ CDM is *false*; extended models may well be true, and adding parameters in response to better data is, sometimes, simply how science advances. It is a prediction about the model’s *mode*: it will continue to meet anomalies by absorbing them, because that is what it has done at every juncture for ninety years, from hot-to-cold dark matter, to the addition of Λ , to feedback tuning, to early dark energy, to dynamical dark energy, to — if it comes to it — a controlled breaking of isotropy.

11. The crux

Philosophers of science have a name for the worry. A *progressive* research program absorbs anomalies while making new, confirmed predictions; a *degenerative* one absorbs them while making none, each fix bought purely to cover a known failure. The Ptolemaic astronomers who added epicycle upon epicycle were not stupid, and their model fit the data superbly to the end. It was simply doing more accommodating than predicting, and a simpler picture was waiting.

The honest position on Λ CDM is that it is *both* — a model with a spine of genuine, risky, confirmed predictions (the CMB peaks, the acoustic ruler) and a long, growing tail of after-the-fact accommodations (the dark-matter temperature, the cosmological constant, the feedback parameters, the early dark energy, the evolving w , perhaps soon the broken symmetry). Which list grows faster, in the decade now opening, is the thing to watch.

It is worth at least noticing that the alternative tradition behaves differently. Modified-dynamics approaches — including the emergent-gravity proposal that ties the single galactic acceleration scale to the cosmological constant, $a_0 = c^2 \sqrt{(\Lambda/32\pi)}$ — earn their keep, when they do, by *prediction*: a fixed number, a falsifiable redshift trend, a solar-system test that has already either passed or failed. Such theories are far less developed than Λ CDM, carry real unsolved problems of their own, and may yet be wrong. But they share a virtue the standard model has slowly traded away: they can be *killed* by a single measurement, because they were not built with the freedom to absorb it.

A model that always fits is comfortable. The cosmic dipole is a reminder that comfort is not the same as truth — and that the measurement which finally refuses to be absorbed, if it comes, will

more likely arrive from a theory that named its number in advance than from one that has spent ninety years learning how to fit whatever the sky sends.

Sources and further reading

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- DESI Collaboration (2024, 2025) — BAO and the preference for evolving dark energy.
- Secrest, N. J. et al. (2021), *ApJL* 908, L51, *A Test of the Cosmological Principle with Quasars* — the WISE quasar dipole, 4.9σ .
- Secrest, N. J. et al. (2022), *ApJL* 937, L31, *A Challenge to the Standard Cosmological Model* — WISE + NVSS, 5.1σ .
- See also the active reassessment of the dipole (Bayesian estimators; CatWISE2020 re-analyses; kinematic number-count contributions, 2023–2025) and recent colloquium-level reviews of the cosmic dipole anomaly.

An interpretive review. Λ CDM’s genuine predictive successes (the CMB acoustic peaks, the baryon-acoustic ruler) are credited in full; the pattern of post-hoc accommodation is documented in full; the cosmic dipole is presented as a sharp but still-contested anomaly, not a settled refutation. The forecast concerns the model’s mode of response, not its truth. The author’s own emergent-gravity proposal is noted as an alternative, with its unsolved problems acknowledged. DOI of the related framework: 10.5281/zenodo.20721540.